

REFERENCES: READ Reactor Theory Manual Chapter 2, Sections §2.1-§2.3
Delayed Neutron Lecture #7 PPT Slides
PLEASE SHOW ALL WORK!

1. **(4 PTS):** If $\ell_p = 3 \times 10^{-5}$ sec, $\ell_d = 12.7$ sec, $\beta = 0.0073$, and $\rho = +200$ pcm, calculate P_f (Watts) at $t = 10$ sec if $P_i = 100$ Watts. **Reference Eq. 2.1 and 2.3 from NE235 Theory Manual**

$$1^{st}: \text{Eq. 2.3: } \bar{l} = l_d(\beta) + l_p(1 - \beta); \text{ so } \bar{l} = 12.7(0.0073) + 3 \times 10^{-5}(1 - \beta) = 0.09274 \text{ sec}$$

$$\text{Where } +200 \text{ pcm} = 0.002 \Delta K/K$$

$$2^{nd}: K_{eff} = \frac{1}{1 - \rho} = \frac{1}{1 - 0.002 \frac{\Delta K}{K}} = 1.002004$$

$$\text{Eq. 2.1: } P_f = P_o e^{\left(\frac{K_{eff} - 1}{\bar{l}}\right)t} = 100 e^{\left(\frac{1.002004 - 1}{0.09274}\right)10} = \mathbf{124.1 \text{ watts}}$$

2. **(2 PTS):** Referencing Fig. 2.2 – Neutron Multiplication Cycle:
- If we adopt the “storage” mentality as a description of delayed neutron fraction, what is the K_{eff} of the cycle if the ratio of neutrons “placed into storage” (β) equals the release rate of delayed neutrons from storage ($\Sigma \lambda_i C_i$)?

$$K_{eff} = 1$$

- Which of the six groups (C_i) of delayed neutron precursors takes the longest to reach saturation activity? What is the average half life of this group?

Group 1 with an average half life of 55.7 sec.

3. **(5 PTS):** As a ‘Light Water Reactor’ (LWR) core like the PULSTAR moves from ‘Beginning of Life’ (BOL) to ‘End of Life’ (EOL) (*mark the following statements True or False*):

T/F?	Statement
T	... the amount of ^{235}U in the LWR fuel decreases.
F	... the amount of ^{239}Pu in the LWR fuel decreases.
T	... the delayed neutron fraction β decreases.
T	... the Startup Rate (DPM) increases for a given net positive reactivity (REF FIG 2.4).
F	... the reactor power will change more slowly for a given net positive reactivity.

CONTROL ROD CALIBRATION LAB QUESTIONS – REFERENCE EXCEL DATA FILE FROM LAB:

1. (2 PTS) Which PULSTAR control rod did we calibrate and which rods did we balance it against to re-establish criticality at each step?

The safety #1 rod, as balanced against Safety #2 and the Reg. Rod.

2. (1 PTS) Which nuclear instrument channel (e.g. Safety, Linear, Log-N or NFM) on the console did we use to measure the doubling time for each pull?

Linear Channel

3. Looking at the excel data file for the lab:

- a) (2 PTS) What was the maximum differential rod worth measured (pcm/in)? At what rod height (in)?

180.6 pcm/inch, if they are using the 1-Group
167.0 pcm/inch, if they are using the Curve
at 12.52"/13.49" (or 13.00" if they average)

- b) (1 PTS) What was the total sum integral rod worth (in pcm) we measured after the last rod pull?

1105.6 pcm, if they are using the 1-Group
1057.0 pcm, if they are using the Curve
(accept either + or -)

- c) (1PTS) Why did the measured differential worth increase initially and then start to decrease?

The differential worth follows the neutron flux gradient inside the core. Initially the bottom of the control rod was below the peak flux. Once the bottom of the rod pulled up past the peak flux and the flux started to decrease, the differential worth decreased.